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Q: A no. is divided by 406 leaves remainder 115, what will be remainder when it will be divided by 29?

Ans.

(M-1)

$$406 \overline{)\, n \,}\ (a$$
$$\underline{-\ \ -}$$
$$115$$

dividend = divisor × quotient + Rem.

$n = 406 \times a + 115$

$n = 406a + 115$

$\div 29$ both side.

$$\frac{n}{29} = \frac{406a}{29} + \frac{115}{29}$$

$n = 0 + 28$

$n = 28$

$$29 \overline{)\, 406a \,}\ (14$$
$$\underline{-\ 406a}$$
$$\times$$
$$\times$$

when n is divided by 29 remainder = 8.

Ans = 28

(M-2)

assume q = 1

$$\begin{array}{r} 1 \\ 406 \overline{)\, n \,} \\ -\ 406 \\ \hline 115 \end{array} \qquad \begin{array}{r} \\ 29 \overline{)\, n \,} \\ \\ ? \end{array}$$

$n - 406 = 115$

$n = 406 + 115$

$n = 521$ (assumed)

$$\begin{array}{r} 17 \\ 29 \overline{)\, 529 \,} \\ -\ 29 \\ \hline 231 \\ -\ 203 \\ \hline 28 \end{array}$$

Ans = 28

formula $\frac{406 + 115}{29} = \text{Rem}$

$\frac{\text{old divisor} + \text{old rem.}}{\text{new divisor}}$

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Q: When a four digit no is divided by 85, it leaves a remainder of 39, if same no. is divided by 17, remainder would be?

A. 85) n (a 17) n (

39 ?

$\frac{85+39}{17} \Rightarrow \frac{124}{17} = 5$

17) 124 (
119
5

or

$\frac{85}{17} = 0 + \frac{39}{17} \Rightarrow 5 \Rightarrow 0+5 = 5$ Ans.

Q: When a no. is divided by 899, it leaves a remainder of 63, if same no. is divided by 29, the remainder would be?

A. $\frac{899+63}{29} \Rightarrow 5$

Q: A no. when divided by D leaves a remainder of 8 and when divided by 3D leaves a remainder of 21. What is the remainder left, when twice no. is divided by 3D?

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Ans.

$D \overline{)n}\,(a$ $\quad$ $3D \overline{)n}\,(b$ $\quad$ $3b \overline{)2n}$

$\quad 8$ $\quad$ 21 $\quad$ 9

$n = D \times a + 8 \rightarrow (1)$

$n = 3D \times b + 21 \rightarrow (2)$

$D \times a + 8 = 3D \times b + 21$

$Da - 3Db = 13$

$D[a - 3b] = 13$

prime no.

Only, $p \times q = 13$

when, $p = 13$, $q = 1$ | $p = 1$, $q = 13$

$D = 1$, $[a - 3b] = 13$ ✗

$D = 13$, $[a - 3b] = 1$ ✓

$D = 13$

$a - 3b = 1$

$a = 4$, $b = 1$ } ans.

$n = Da + 8$

$n = 13 \times 4 + 8$

$n = 60$ Assumed (not fixed)

$$\begin{array}{r} 3 \\ 39 \overline{)120}\,(\\ 117 \\ \hline 3 \end{array}$$

$b = 3$

$3D = 39$

$2n = 120$

As above we said, $D \div n = a$ quotient & 8 rem.

अगर 2 से ÷ करेंगे तो 8 नहीं 0 rem. आएगा।

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Q: If $22^3 + 23^3 + 24^3 + \cdots + 87^3 + 88^3$ is divided by 110, then remainder will be?

Ans

$\Rightarrow \frac{a+b}{a+b} \Rightarrow$ Rem = 0

$\Rightarrow \frac{a^3+b^3}{a+b} \Rightarrow$ Rem = 0

$\downarrow$

$a^3+b^3+3ab(a+b)$

$\frac{a^3+b^3}{a+b} \Rightarrow \frac{(a+b)^3 - 3ab(a+b)}{a+b}$

$(a+b)^2 - 3ab$ (after div is no remain)

* $\frac{a^n+b^n}{a+b}$ n = odd, rem = 0

$\downarrow$

$\frac{a^n+b^n+c^n+d^n+\cdots}{a+b+c+d}$ n = odd, rem = 0

$22^3+23^3+24^3+ \cdots 87^3+88^3$

$\Rightarrow \frac{22^3+88^3}{\text{rem} = 110} + \frac{23^3+87^3}{\text{rem} = 110} + \cdots$

$\frac{22^3+88^3}{110} + \frac{23^3+87^3}{110} + \cdots$

$\frac{22^3+88^3}{22+88} + \frac{23^3+87^3}{23+87} - - -$

Remainder = 0

$(22-54)$ (55) $(56-88)$

pair single pair

66 odd pair

odd

22 23 24 ... 87 88

$a = 22$, $n = ?$

$d = 1$

$a_n = a + (n-1)d$

$88 = 22 + (n-1) \times 1$

$66 = n-1$

$n = 67$

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$= 0 + 0 - - - + 0 + \frac{55^3}{110} = \frac{\cancel{55} \times 55 \times 55}{\cancel{110}_{2}}$

$= \frac{55 \times 55}{2}$ → either rem = 0, rem = 1

$= \frac{\overset{\text{unit}}{5}}{2}$ → (1) rem.

Ans = 1

Q: If $16^7 + 17^7 + 18^7 + 19^7$ is divided by 5, 7, 14 & 35 and remainder thus obtained are R_1, R_2, R_3 & R_4 respectively, find $R_1 + R_2 + R_3 + R_4$.

Ans $\frac{16^7 + 17^7 + 18^7 + 19^7}{16+17+18+19} = \frac{16^7 + 17^7 + 18^7 + 19^7}{70} = R_{rem}$

* Property :-

$12 = \{1, 2, 3, 4, 6, 12\}$ factors

$\frac{n}{12} = ✓$, then $\frac{n}{6}, \frac{n}{3}, \frac{n}{4}, \frac{n}{2}, \frac{n}{1}\}$

अर्थात् n 12 से ÷ होगा तो 12 के factors सभी होंगे।

$70 = \{1, 5, 7, 14, 35\}$ ✓

factors.

$\frac{n}{5}$ ✓, $\frac{n}{7}$ ✓, $\frac{n}{14}$, $\frac{n}{35}$ ✓

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$R_1 + R_2 + R_3 + R_4 = 0$

Q: If 'n' is a natural no such that $10^{100} < n < 10^{101}$ & sum of digits of n is 2, then no. of values 'n' can take is?

a) 74 b) 89 c) 100 d) 101

Ans.

Case i) If $10^1 < n\ 10^2$

1 1
2 0 } 2 no. can be formed

ii) If $10^2 < 10^3$

1 0 1
1 1 0
2 0 0 } 3

iii) $10^3 < 10^4$

1 0 0 1
1 1 0 0
1 0 1 0
2 0 0 0 } 4

(iv) $10^{100} < n < 10^{101}$

- - - } 101 Ans.
- - -